PV Soiling Analysis Report

Experiment Details

Field	Value
Experiment	Minna Soiling Test 2026

Weekly Measurements

Week	Temp (°C)	Clean P (W)	Dusty P (W)	Soiling Ratio	Soiling Loss (%)
1	29.0	578.35	572.97	0.9907	0.93
2	29.0	578.64	561.38	0.9702	2.98
3	29.0	578.02	550.44	0.9523	4.77
4	29.0	578.13	539.50	0.9332	6.68
5	29.0	578.11	530.06	0.9169	8.31

Regression Analysis

Parameter	Value
Slope	1.846 % per week
Intercept	-0.802
R^2 Value	0.999
Equation	SL = 1.846 × Week + -0.802

Summary and Recommendations

Key Findings:
• Week 1 Soiling Loss: 0.93%
• Week 5 Soiling Loss: 8.31%
• Total Increase: 7.38%
• Average Soiling Rate: 1.85% per week

Recommendation:
Based on the observed soiling trend, cleaning every 2-3 weeks is recommended.

During Harmattan conditions, more frequent cleaning may be necessary.

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